

Production Functions

The principal activity of any firm is to turn inputs into outputs. Because economists are interested in the choices the firm makes in accomplishing this goal, but wish to avoid discussing many of the engineering intricacies involved, they have chosen to construct an abstract model of production. In this model the relationship between inputs and outputs is formalized by a *production function* of the form

$$q = f(k, l, m, \dots), \quad (9.1)$$

where q represents the firm's output of a particular good during a period,¹ k represents the machine (i.e., capital) usage during the period, l represents hours of labor input, m represents raw materials used,² and the notation indicates the possibility of other variables affecting the production process. Equation 9.1 is assumed to provide, for any conceivable set of inputs, the engineer's solution to the problem of how best to combine those inputs to get output.

MARGINAL PRODUCTIVITY

In this section we look at the change in output brought about by a change in one of the productive inputs. For the purposes of this examination (and indeed for most of the purposes of this book), it will be more convenient to use a simplified production function defined as follows.

DEFINITION

Production function. The firm's *production function* for a particular good, q ,

$$q = f(k, l), \quad (9.2)$$

shows the maximum amount of the good that can be produced using alternative combinations of capital (k) and labor (l).

Of course, most of our analysis will hold for any two inputs to the production process we might wish to examine. The terms *capital* and *labor* are used only for convenience. Similarly, it would be a simple matter to generalize our discussion to cases involving

¹Here we use a lowercase q to represent one firm's output. We reserve the uppercase Q to represent total output in a market. Generally, we assume that a firm produces only one output. Issues that arise in multiproduct firms are discussed in a few footnotes and problems.

²In empirical work, raw material inputs often are disregarded, and output, q , is measured in terms of "value added."

more than two inputs; occasionally, we will do so. For the most part, however, limiting the discussion to two inputs will be helpful because we can show these inputs on two-dimensional graphs.

Marginal physical product

To study variation in a single input, we define marginal physical product as follows.

DEFINITION

Marginal physical product. The *marginal physical product* of an input is the additional output that can be produced by using one more unit of that input while holding all other inputs constant. Mathematically,

$$\begin{aligned}\text{marginal physical product of capital} &= MP_k = \frac{\partial q}{\partial k} = f_k, \\ \text{marginal physical product of labor} &= MP_l = \frac{\partial q}{\partial l} = f_l.\end{aligned}\tag{9.3}$$

Notice that the mathematical definitions of marginal product use partial derivatives, thereby properly reflecting the fact that all other input usage is held constant while the input of interest is being varied. For example, consider a farmer hiring one more laborer to harvest the crop but holding all other inputs constant. The extra output this laborer produces is that farmhand's marginal physical product, measured in physical quantities, such as bushels of wheat, crates of oranges, or heads of lettuce. We might observe, for example, that 50 workers on a farm are able to produce 100 bushels of wheat per year, whereas 51 workers, with the same land and equipment, can produce 102 bushels. The marginal physical product of the 51st worker is then 2 bushels per year.

Diminishing marginal productivity

We might expect that the marginal physical product of an input depends on how much of that input is used. Labor, for example, cannot be added indefinitely to a given field (while keeping the amount of equipment, fertilizer, and so forth fixed) without eventually exhibiting some deterioration in its productivity. Mathematically, the assumption of diminishing marginal physical productivity is an assumption about the second-order partial derivatives of the production function:

$$\begin{aligned}\frac{\partial MP_k}{\partial k} &= \frac{\partial^2 f}{\partial k^2} = f_{kk} = f_{11} < 0, \\ \frac{\partial MP_l}{\partial l} &= \frac{\partial^2 f}{\partial l^2} = f_{ll} = f_{22} < 0.\end{aligned}\tag{9.4}$$

The assumption of diminishing marginal productivity was originally proposed by the nineteenth-century economist Thomas Malthus, who worried that rapid increases in population would result in lower labor productivity. His gloomy predictions for the future of humanity led economics to be called the “dismal science.” But the mathematics of the production function suggests that such gloom may be misplaced. Changes in the marginal productivity of labor over time depend not only on how labor input is growing but also on changes in other inputs, such as capital. That is, we must also be concerned with $\partial MP_l / \partial k = f_{lk}$. In most cases, $f_{lk} > 0$, thus, declining labor productivity as *both* l and k increase is not a foregone conclusion. Indeed, it appears that labor productivity has risen significantly since Malthus' time, primarily because increases in capital inputs (along with technical improvements) have offset the impact of decreasing marginal productivity alone.

Average physical productivity

In common usage, the term *labor productivity* often means *average productivity*. When it is said that a certain industry has experienced productivity increases, this is taken to mean that output per unit of labor input has increased. Although the concept of average productivity is not nearly as important in theoretical economic discussions as marginal productivity is, it receives a great deal of attention in empirical discussions. Because average productivity is easily measured (say, as so many bushels of wheat per labor-hour input), it is often used as a measure of efficiency. We define the average product of labor (AP_l) to be

$$AP_l = \frac{\text{output}}{\text{labor input}} = \frac{q}{l} = \frac{f(k, l)}{l}. \quad (9.5)$$

Notice that AP_l also depends on the level of capital used. This observation will prove to be important when we examine the measurement of technical change at the end of this chapter.

EXAMPLE 9.1 A Two-Input Production Function

Suppose the production function for flyswatters during a particular period can be represented by

$$q = f(k, l) = 600k^2l^2 - k^3l^3. \quad (9.6)$$

To construct the marginal and average productivity functions of labor (l) for this function, we must assume a particular value for the other input, capital (k). Suppose $k = 10$. Then the production function is given by

$$q = 60,000l^2 - 1,000l^3. \quad (9.7)$$

Marginal product. The marginal productivity function (when $k = 10$) is given by

$$MP_l = \frac{\partial q}{\partial l} = 120,000l - 3,000l^2, \quad (9.8)$$

which diminishes as l increases, eventually becoming negative. This implies that q reaches a maximum value. Setting MP_l equal to 0,

$$120,000l - 3,000l^2 = 0 \quad (9.9)$$

yields

$$40l = l^2 \quad (9.10)$$

or

$$l = 40 \quad (9.11)$$

as the point at which q reaches its maximum value. Labor input beyond 40 units per period actually reduces total output. For example, when $l = 40$, Equation 9.7 shows that $q = 32$ million flyswatters, whereas when $l = 50$, production of flyswatters amounts to only 25 million.

Average product. To find the average productivity of labor in flyswatter production, we divide q by l , still holding $k = 10$:

$$AP_l = \frac{q}{l} = 60,000l - 1,000l^2. \quad (9.12)$$

Again, this is an inverted parabola that reaches its maximum value when

$$\frac{\partial AP_l}{\partial l} = 60,000 - 2,000l = 0, \quad (9.13)$$

which occurs when $l = 30$. At this value for labor input, Equation 9.12 shows that $AP_l = 900,000$, and Equation 9.8 shows that MP_l is also 900,000. When AP_l is at a maximum, average and marginal productivities of labor are equal.³

Notice the relationship between total output and average productivity that is illustrated by this example. Even though total production of flyswatters is greater with 40 workers (32 million) than with 30 workers (27 million), output per worker is higher in the second case. With 40 workers, each worker produces 800,000 flyswatters per period, whereas with 30 workers each worker produces 900,000. Because capital input (flyswatter presses) is held constant in this definition of productivity, the diminishing marginal productivity of labor eventually results in a declining level of output per worker.

QUERY: How would an increase in k from 10 to 11 affect the MP_l and AP_l functions here? Explain your results intuitively.

ISOQUANT MAPS AND THE RATE OF TECHNICAL SUBSTITUTION

To illustrate possible substitution of one input for another in a production function, we use its *isoquant map*. Again, we study a production function of the form $q = f(k, l)$, with the understanding that “capital” and “labor” are simply convenient examples of any two inputs that might happen to be of interest. An isoquant (from *iso*, meaning “equal”) records those combinations of k and l that are able to produce a given quantity of output. For example, all those combinations of k and l that fall on the curve labeled “ $q = 10$ ” in Figure 9.1 are capable of producing 10 units of output per period. This isoquant then records the fact that there are many alternative ways of producing 10 units of output. One way might be represented by point A: We would use l_A and k_A to produce 10 units of output. Alternatively, we might prefer to use relatively less capital and more labor and therefore would choose a point such as B. Hence we may define an isoquant as follows.

DEFINITION

Isoquant. An *isoquant* shows those combinations of k and l that can produce a given level of output (say, q_0). Mathematically, an isoquant records the set of k and l that satisfies

$$f(k, l) = q_0. \quad (9.14)$$

As was the case for indifference curves, there are infinitely many isoquants in the k - l plane. Each isoquant represents a different level of output. Isoquants record successively higher levels of output as we move in a northeasterly direction. Presumably, using more

³This result is general. Because

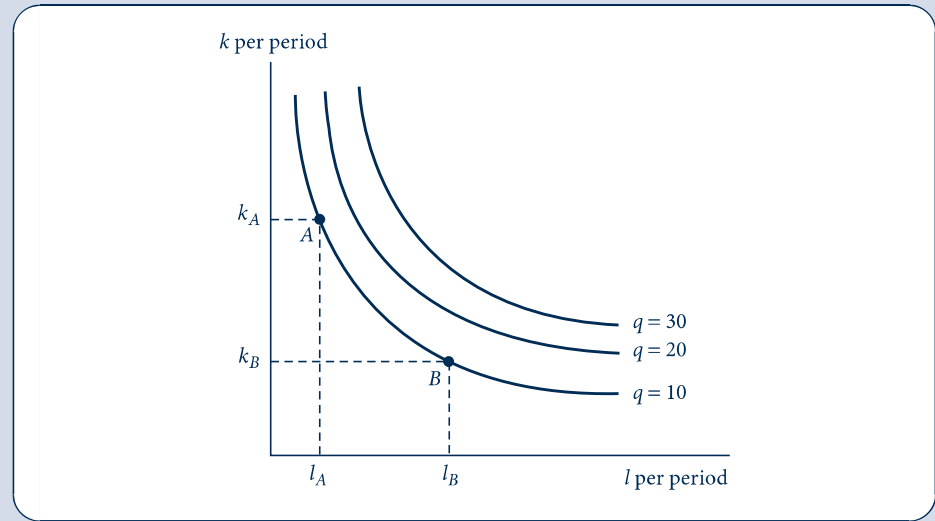
$$\frac{\partial AP_l}{\partial l} = \frac{l \cdot MP_l - q}{l^2},$$

at a maximum $l \cdot MP_l = q$ or $MP_l = AP_l$.

FIGURE 9.1

An Isoquant Map

Isoquants record the alternative combinations of inputs that can be used to produce a given level of output. The slope of these curves shows the rate at which l can be substituted for k while keeping output constant. The negative of this slope is called the (marginal) *rate of technical substitution* (RTS). In the figure, the RTS is positive and diminishing for increasing inputs of labor.



of each of the inputs will permit output to increase. Two other isoquants (for $q = 20$ and $q = 30$) are shown in Figure 9.1. You will notice the similarity between an isoquant map and the individual's indifference curve map discussed in Part 2. They are indeed similar concepts because both represent “contour” maps of a particular function. For isoquants, however, the labeling of the curves is measurable—an output of 10 units per period has a quantifiable meaning. Therefore, economists are more interested in studying the shape of production functions than in examining the exact shape of utility functions.

The marginal rate of technical substitution (RTS)

The slope of an isoquant shows how one input can be traded for another while holding output constant. Examining the slope provides information about the technical possibility of substituting labor for capital. A formal definition follows.

DEFINITION

Marginal rate of technical substitution. The *marginal rate of technical substitution* (RTS) shows the rate at which labor can be substituted for capital while holding output constant along an isoquant. In mathematical terms,

$$RTS (l \text{ for } k) = - \left. \frac{dk}{dl} \right|_{q=q_0}. \quad (9.15)$$

In this definition, the notation is intended as a reminder that output is to be held constant as l is substituted for k . The particular value of this trade-off rate will depend not only on the level of output but also on the quantities of capital and labor being used. Its value depends on the point on the isoquant map at which the slope is to be measured.

RTS and marginal productivities

To examine the shape of production function isoquants, it is useful to prove the following result: The *RTS* (of l for k) is equal to the ratio of the marginal physical productivity of labor (MP_l) to the marginal physical productivity of capital (MP_k). Imagine using Equation 9.14 to graph the q_0 isoquant. We would substitute a sequence of increasing values of l and see how k would have to adjust to keep output constant at q_0 . The graph of the isoquant is really the graph of the implicit function $k(l)$ satisfying

$$q_0 = f(k(l), l). \quad (9.16)$$

Just as we did in the section on implicit functions in Chapter 2 (see in particular Equation 2.22), we can use the chain rule to differentiate Equation 9.16, giving

$$0 = f_k \frac{dk}{dl} + f_l = MP_k \frac{dk}{dl} + MP_l, \quad (9.17)$$

where the initial 0 appears because q_0 is being held constant; therefore, the derivative of the left side of Equation 9.16 with respect to l equals 0. Rearranging Equation 9.17 gives

$$RTS \text{ (} l \text{ for } k \text{)} = - \left. \frac{dk}{dl} \right|_{q=q_0} = \frac{MP_l}{MP_k}. \quad (9.18)$$

Hence the *RTS* is given by the ratio of the inputs' marginal productivities.

Equation 9.18 shows that those isoquants that we actually observe must be negatively sloped. Because both MP_l and MP_k will be non-negative (no firm would choose to use a costly input that reduced output), the *RTS* also will be positive (or perhaps zero). Because the slope of an isoquant is the negative of the *RTS*, any firm we observe will not be operating on the positively sloped portion of an isoquant. Although it is mathematically possible to devise production functions whose isoquants have positive slopes at some points, it would not make economic sense for a firm to opt for such input choices.

Reasons for a diminishing *RTS*

The isoquants in Figure 9.1 are drawn not only with a negative slope (as they should be) but also as convex curves. Along any one of the curves, the *RTS* is *diminishing*. For high ratios of k to l , the *RTS* is a large positive number, indicating that a great deal of capital can be given up if one more unit of labor becomes available. On the other hand, when a lot of labor is already being used, the *RTS* is low, signifying that only a small amount of capital can be traded for an additional unit of labor if output is to be held constant. This assumption would seem to have some relationship to the assumption of diminishing marginal productivity. A hasty use of Equation 9.18 might lead one to conclude that an increase in l accompanied by a decrease in k would result in a decrease in MP_l , an increase in MP_k , and, therefore, a decrease in the *RTS*. The problem with this quick "proof" is that the marginal productivity of an input depends on the level of *both* inputs—changes in l affect MP_k and vice versa. It is not possible to derive a diminishing *RTS* from the assumption of diminishing marginal productivity alone.

To see why this is so mathematically, assume that $q = f(k, l)$ and that f_k and f_l are positive (i.e., the marginal productivities are positive). Assume also that $f_{kk} < 0$ and $f_{ll} < 0$ (that the marginal productivities are diminishing). To show that isoquants are convex, we would like to show that $d(RTS)/dl < 0$. Because $RTS = f_l/f_k$, we have

$$\frac{dRTS}{dl} = \frac{d(f_l/f_k)}{dl}. \quad (9.19)$$

Because f_l and f_k are functions of both k and l , we must be careful in taking the derivative of this expression:

$$\frac{dRTS}{dl} = \frac{f_k(f_{ll} + f_{lk} \cdot dk/dl) - f_l(f_{kl} + f_{kk} \cdot dk/dl)}{(f_k)^2}. \quad (9.20)$$

Using the fact that $dk/dl = -f_l/f_k$ along an isoquant and Young's theorem ($f_{kl} = f_{lk}$), we have

$$\frac{dRTS}{dl} = \frac{f_k^2 f_{ll} - 2f_k f_l f_{kl} + f_l^2 f_{kk}}{(f_k)^3}. \quad (9.21)$$

Because we have assumed $f_k > 0$, the denominator of this function is positive. Hence the whole fraction will be negative if the numerator is negative. Because f_{ll} and f_{kk} are both assumed to be negative, the numerator definitely will be negative if f_{kl} is positive. If we can assume this, we have shown that $dRTS/dl < 0$ (that the isoquants are convex).⁴

Importance of cross-productivity effects

Intuitively, it seems reasonable that the cross-partial derivative $f_{kl} = f_{lk}$ should be positive. If workers had more capital, they would have higher marginal productivities. Although this is probably the most prevalent case, it does not necessarily have to be so. Some production functions have $f_{kl} < 0$, at least for a range of input values. When we assume a diminishing RTS (as we will throughout most of our discussion), we are therefore making a stronger assumption than simply diminishing marginal productivities for each input—specifically, we are assuming that marginal productivities diminish “rapidly enough” to compensate for any possible negative cross-productivity effects. Of course, as we shall see later, with three or more inputs, things become even more complicated.

EXAMPLE 9.2 A Diminishing RTS

In Example 9.1, the production function for flyswatters was given by

$$q = f(k, l) = 600k^2l^2 - k^3l^3. \quad (9.22)$$

General marginal productivity functions for this production function are

$$\begin{aligned} MP_l = f_l &= \frac{\partial q}{\partial l} = 1,200k^2l - 3k^3l^2, \\ MP_k = f_k &= \frac{\partial q}{\partial k} = 1,200kl^2 - 3k^2l^3. \end{aligned} \quad (9.23)$$

Notice that each of these depends on the values of both inputs. Simple factoring shows that these marginal productivities will be positive for values of k and l for which $kl < 400$.

Because

$$f_{ll} = 1,200k^2 - 6k^3l$$

and

$$f_{kk} = 1,200l^2 - 6kl^3, \quad (9.24)$$

it is clear that this function exhibits diminishing marginal productivities for sufficiently large values of k and l . Indeed, again by factoring each expression, it is easy to show that $f_{ll}, f_{kk} < 0$ if

⁴As we pointed out in Chapter 2, functions for which the numerator in Equation 9.21 is negative are called (strictly) *quasi-concave functions*.

$kl > 200$. However, even within the range $200 < kl < 400$ where the marginal productivity relations for this function behave “normally,” this production function may not necessarily have a diminishing *RTS*. Cross-differentiation of either of the marginal productivity functions (Equation 9.23) yields

$$f_{kl} = f_{lk} = 2,400kl - 9k^2l^2, \quad (9.25)$$

which is positive only for $kl < 266$.

Therefore, the numerator of Equation 9.21 will definitely be negative for $200 < kl < 266$, but for larger-scale flyswatter factories the case is not so clear because f_{kl} is negative. When f_{kl} is negative, increases in labor input reduce the marginal productivity of capital. Hence the intuitive argument that the assumption of diminishing marginal productivities yields an unambiguous prediction about what will happen to the *RTS* ($= f_l/f_k$) as l increases and k decreases is incorrect. It all depends on the relative effects on marginal productivities of diminishing marginal productivities (which tend to reduce f_l and increase f_k) and the contrary effects of cross-marginal productivities (which tend to increase f_l and reduce f_k). Still, for this flyswatter case, it is true that the *RTS* is diminishing throughout the range of k and l where marginal productivities are positive. For cases where $266 < kl < 400$, the diminishing marginal productivities exhibited by the function are sufficient to overcome the influence of a negative value for f_{kl} on the convexity of isoquants.

QUERY: For cases where $k = l$, what can be said about the marginal productivities of this production function? How would this simplify the numerator for Equation 9.21? How does this permit you to more easily evaluate this expression for some larger values of k and l ?

RETURNS TO SCALE

We now proceed to characterize production functions. A first question that might be asked about them is how output responds to increases in all inputs together. For example, suppose that all inputs were doubled: Would output double or would the relationship not be so simple? This is a question of the *returns to scale* exhibited by the production function that has been of interest to economists ever since Adam Smith intensively studied the production of pins. Smith identified two forces that came into operation when the conceptual experiment of doubling all inputs was performed. First, a doubling of scale permits a greater division of labor and specialization of function. Hence there is some presumption that efficiency might increase—production might more than double. Second, doubling of the inputs also entails some loss in efficiency because managerial over-seeing may become more difficult given the larger scale of the firm. Which of these two tendencies will have a greater effect is an important empirical question.

These concepts can be defined technically as follows:

DEFINITION

Returns to scale. If the production function is given by $q = f(k, l)$ and if all inputs are multiplied by the same positive constant t (where $t > 1$), then we classify the *returns to scale* of the production function by

Effect on Output	Returns to Scale
$f(tk, tl) = tf(k, l) = tq$	Constant
$f(tk, tl) < tf(k, l) = tq$	Decreasing
$f(tk, tl) > tf(k, l) = tq$	Increasing

In intuitive terms, if a proportionate increase in inputs increases output by the same proportion, the production function exhibits constant returns to scale. If output increases less than proportionately, the function exhibits diminishing returns to scale. And if output increases more than proportionately, there are increasing returns to scale. As we shall see, it is theoretically possible for a function to exhibit constant returns to scale for some levels of input usage and increasing or decreasing returns for other levels.⁵ Often, however, economists refer to returns to scale of a production function with the implicit understanding that only a fairly narrow range of variation in input usage and the related level of output is being considered.

Constant returns to scale

There are economic reasons why a firm's production function might exhibit constant returns to scale. If the firm operates many identical plants, it may increase or decrease production simply by varying the number of them in current operation. That is, the firm can double output by doubling the number of plants it operates, and that will require it to employ precisely twice as many inputs. Empirical studies of production functions often find that returns to scale are roughly constant for the firms studied (at least around for outputs close to the firms' established operating levels—the firms may exhibit increasing returns to scale as they expand to their established size). For all these reasons, the constant returns-to-scale case seems worth examining in somewhat more detail.

When a production function exhibits constant returns to scale, it meets the definition of “homogeneity” that we introduced in Chapter 2. That is, the production is homogeneous of degree 1 in its inputs because

$$f(tk, tl) = t^1 f(k, l) = tq. \quad (9.26)$$

In Chapter 2 we showed that, if a function is homogeneous of degree k , its derivatives are homogeneous of degree $k - 1$. In this context this implies that the marginal productivity functions derived from a constant returns-to-scale production function are homogeneous of degree 0. That is,

$$\begin{aligned} MP_k &= \frac{\partial f(k, l)}{\partial k} = \frac{\partial f(tk, tl)}{\partial k}, \\ MP_l &= \frac{\partial f(k, l)}{\partial l} = \frac{\partial f(tk, tl)}{\partial l} \end{aligned} \quad (9.27)$$

for any $t > 0$. In particular, we can let $t = 1/l$ in Equations 9.27 and get

$$\begin{aligned} MP_k &= \frac{\partial f(k/l, 1)}{\partial k}, \\ MP_l &= \frac{\partial f(k/l, 1)}{\partial l}. \end{aligned} \quad (9.28)$$

That is, the marginal productivity of any input depends only on the ratio of capital to labor input, not on the absolute levels of these inputs. This fact is especially important, for example, in explaining differences in productivity among industries or across countries.

⁵A local measure of returns to scale is provided by the scale elasticity, defined as

$$e_{q,t} = \frac{\partial f(tk, tl)}{\partial t} \cdot \frac{t}{f(tk, tl)},$$

where this expression is to be evaluated at $t = 1$. This parameter can, in principle, take on different values depending on the level of input usage. For some examples using this concept, see Problem 9.9.

Homothetic production functions

One consequence of Equations 9.28 is that the RTS ($= MP_l/MP_k$) for any constant returns-to-scale production function will depend only on the ratio of the inputs, not on their absolute levels. That is, such a function will be homothetic (see Chapter 2)—its isoquants will be radial expansions of one another. This situation is shown in Figure 9.2. Along any ray through the origin (where the ratio k/l does not change), the slopes of successively higher isoquants are identical. This property of the isoquant map will be useful to us on several occasions.

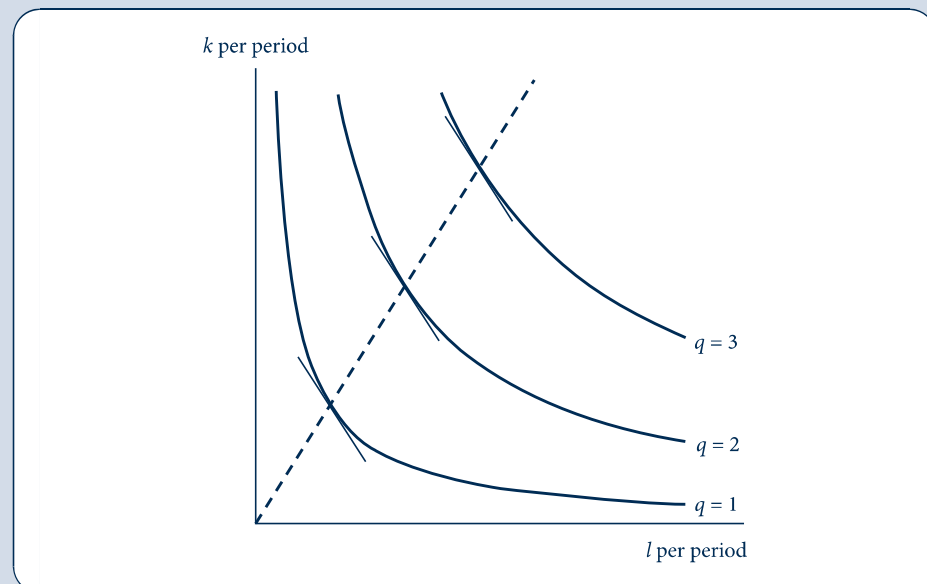
A simple numerical example may provide some intuition about this result. Suppose a large bread order (consisting of, say, 200 loaves) can be filled in one day by three bakers working with three ovens or by two bakers working with four ovens. Therefore, the RTS of ovens for bakers is one for one—one extra oven can be substituted for one baker. If this production process exhibits constant returns to scale, two large bread orders (totaling 400 loaves) can be filled in one day, either by six bakers with six ovens or by four bakers with eight ovens. In the latter case, two ovens are substituted for two bakers, so again the RTS is one for one. In constant returns-to-scale cases, expanding the level of production does not alter trade-offs among inputs; thus, production functions are homothetic.

A production function can have a homothetic indifference curve map even if it does not exhibit constant returns to scale. As we showed in Chapter 2, this property of homotheticity is retained by any monotonic transformation of a homogeneous function. Hence increasing or decreasing returns to scale can be incorporated into a constant returns-to-scale function through an appropriate transformation. Perhaps the most common such transformation is exponential. Thus, if $f(k, l)$ is a constant returns-to-scale production

FIGURE 9.2

Isoquant Map for a Constant Returns-to-Scale Production Function

Because a constant returns-to-scale production function is homothetic, the RTS depends only on the ratio of k to l , not on the scale of production. Consequently, along any ray through the origin (a ray of constant k/l), the RTS will be the same on all isoquants. An additional feature is that the isoquant labels increase proportionately with the inputs.



function, we can let

$$F(k, l) = [f(k, l)]^\gamma, \quad (9.29)$$

where γ is any positive exponent. If $\gamma > 1$, then

$$F(tk, tl) = [f(tk, tl)]^\gamma = [tf(k, l)]^\gamma = t^\gamma [f(k, l)]^\gamma = t^\gamma F(k, l) > tF(k, l) \quad (9.30)$$

for any $t > 1$. Hence this transformed production function exhibits increasing returns to scale. The exponent γ captures the *degree* of the increasing returns to scale. A doubling of inputs would lead to a four-fold increase in output if $\gamma = 2$ but an eight-fold increase if $\gamma = 3$. An identical proof shows that the function F exhibits decreasing returns to scale for $\gamma < 1$. Because this function remains homothetic through all such transformations, we have shown that there are important cases where the issue of returns to scale can be separated from issues involving the shape of an isoquant. In these cases, changes in the returns to scale will just change the labels on the isoquants rather than their shapes. In the next section, we will look at how shapes of isoquants can be described.

The n -input case

The definition of returns to scale can be easily generalized to a production function with n inputs. If that production function is given by

$$q = f(x_1, x_2, \dots, x_n) \quad (9.31)$$

and if all inputs are multiplied by $t > 1$, we have

$$f(tx_1, tx_2, \dots, tx_n) = t^k f(x_1, x_2, \dots, x_n) = t^k q \quad (9.32)$$

for some constant k . If $k = 1$, the production function exhibits constant returns to scale. Decreasing and increasing returns to scale correspond to the cases $k < 1$ and $k > 1$, respectively.

The crucial part of this mathematical definition is the requirement that all inputs be increased by the same proportion, t . In many real-world production processes, this provision may make little economic sense. For example, a firm may have only one “boss,” and that number would not necessarily be doubled even if all other inputs were. Or the output of a farm may depend on the fertility of the soil. It may not be literally possible to double the acres planted while maintaining fertility because the new land may not be as good as that already under cultivation. Hence some inputs may have to be fixed (or at least imperfectly variable) for most practical purposes. In such cases, some degree of diminishing productivity (a result of increasing employment of variable inputs) seems likely, although this cannot properly be called “diminishing returns to scale” because of the presence of inputs that are held fixed.

THE ELASTICITY OF SUBSTITUTION

Another important characteristic of the production function is how “easy” it is to substitute one input for another. This is a question about the shape of a single isoquant rather than about the whole isoquant map. Along one isoquant, the rate of technical substitution will decrease as the capital–labor ratio decreases (i.e., as k/l decreases); now we wish to define some parameter that measures this degree of responsiveness. If the *RTS* does not change at all for changes in k/l , we might say that substitution is easy because the ratio of the marginal productivities of the two inputs does not change as the input mix changes. Alternatively, if the *RTS* changes rapidly for small changes in k/l , we would say that

substitution is difficult because minor variations in the input mix will have a substantial effect on the inputs' relative productivities. A scale-free measure of this responsiveness is provided by the *elasticity of substitution*, a concept we encountered informally in our discussion of CES utility functions. Here we will work on providing a more formal definition.

For discrete changes, the elasticity of substitution is given by

$$\sigma = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{\Delta(k/l)}{k/l} \div \frac{\Delta RTS}{RTS} = \frac{\Delta(k/l)}{\Delta RTS} \cdot \frac{RTS}{(k/l)}. \quad (9.33)$$

More often, we will be interested in considering small changes; therefore, a modification of Equation 9.33 will be of more interest:

$$\sigma = \frac{d(k/l)}{d RTS} \cdot \frac{RTS}{k/l} = \frac{d \ln(k/l)}{d \ln RTS}. \quad (9.34)$$

The logarithmic expression follows from mathematical derivations along the lines of Example 2.2 from Chapter 2. All these equations can be collected in the following formal definition.

DEFINITION

Elasticity of substitution. For the production function $q = f(k, l)$, the *elasticity of substitution* (σ) measures the proportionate change in k/l relative to the proportionate change in the RTS along an isoquant. That is,

$$\sigma = \frac{\text{percent } \Delta(k/l)}{\text{percent } \Delta RTS} = \frac{d(k/l)}{d RTS} \cdot \frac{RTS}{k/l} = \frac{d \ln(k/l)}{d \ln RTS} = \frac{d \ln(k/l)}{d \ln(f_l/f_k)}. \quad (9.35)$$

Because along an isoquant k/l and RTS move in the same direction, the value of σ is always positive. Graphically, this concept is illustrated in Figure 9.3 as a movement from point A to point B on an isoquant. In this movement, both the RTS and the ratio k/l will change; we are interested in the relative magnitude of these changes. If σ is high, then the RTS will not change much relative to k/l and the isoquant will be close to linear. On the other hand, a low value of σ implies a rather sharply curved isoquant; the RTS will change by a substantial amount as k/l changes. In general, it is possible that the elasticity of substitution will vary as one moves along an isoquant and as the scale of production changes. Often, however, it is convenient to assume that σ is constant along an isoquant. If the production function is also homothetic, then—because all the isoquants are merely radial blowups— σ will be the same along all isoquants. We will encounter such functions later in this chapter and in many of the end of chapter problems.⁶

The n -input case

Generalizing the elasticity of substitution to the many-input case raises several complications. One approach is to adopt a definition analogous to Equation 9.35; that is, to define

⁶The elasticity of substitution can be phrased directly in terms of the production function and its derivatives in the constant returns-to-scale case as

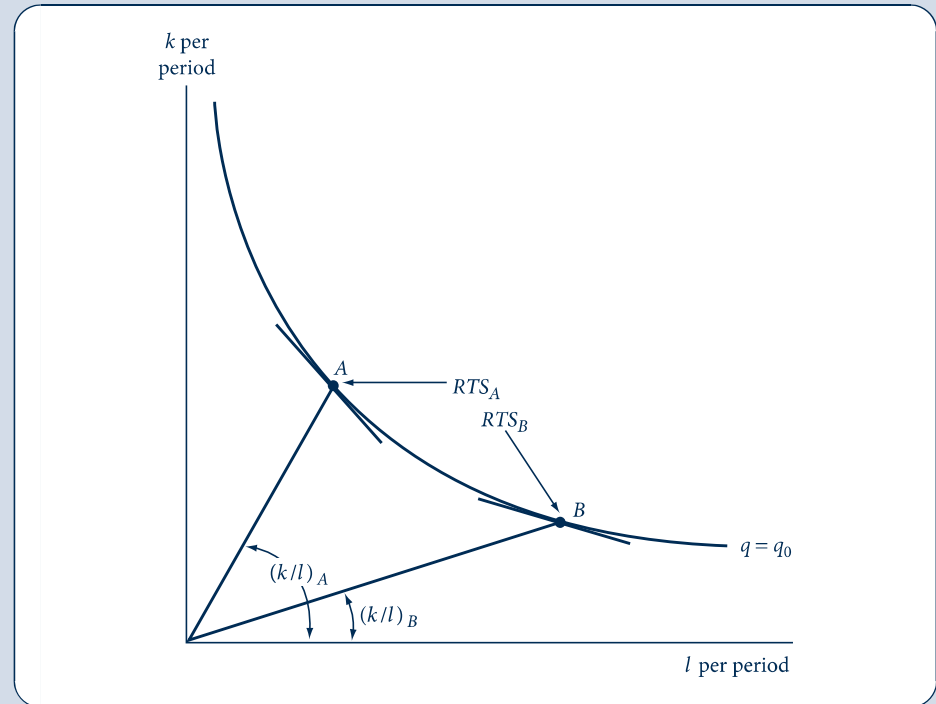
$$\sigma = \frac{f_k \cdot f_l}{f \cdot f_{k,l}}$$

But this form is cumbersome. Hence usually the logarithmic definition in Equation 9.35 is easiest to apply. For a compact summary, see P. Berck and K. Sydsæter, *Economist's Mathematical Manual* (Berlin, Germany: Springer-Verlag, 1999), chap. 5.

FIGURE 9.3

Graphic Description
of the Elasticity of
Substitution

In moving from point A to point B on the $q = q_0$ isoquant, both the capital–labor ratio (k/l) and the RTS will change. The elasticity of substitution (σ) is defined to be the ratio of these proportional changes; it is a measure of how curved the isoquant is.



the elasticity of substitution between two inputs to be the proportionate change in the ratio of the two inputs to the proportionate change in the RTS between them while holding output constant.⁷ To make this definition complete, it is necessary to require that all inputs other than the two being examined be held constant. However, this latter requirement (which is not relevant when there are only two inputs) restricts the value of this potential definition. In real-world production processes, it is likely that any change in the ratio of two inputs will also be accompanied by changes in the levels of other inputs. Some of these other inputs may be complementary with the ones being changed, whereas others may be substitutes, and to hold them constant creates a rather artificial restriction. For this reason, an alternative definition of the elasticity of substitution that permits such complementarity and substitutability in the firm's cost function is generally used in the n -good case. Because this concept is usually measured using cost functions, we will describe it in the next chapter.

⁷That is, the elasticity of substitution between input i and input j might be defined as

$$\sigma_{ij} = \frac{\partial \ln(x_i/x_j)}{\partial \ln(f_j/f_i)}$$

for movements along $f(x_1, x_2, \dots, x_n) = q_0$. Notice that the use of partial derivatives in this definition effectively requires that all inputs other than i and j be held constant when considering movements along the q_0 isoquant.

FOUR SIMPLE PRODUCTION FUNCTIONS

In this section we illustrate four simple production functions, each characterized by a different elasticity of substitution. These are shown only for the case of two inputs, but generalization to many inputs is easily accomplished (see the Extensions for this chapter).

Case 1: Linear ($\sigma = \infty$)

Suppose that the production function is given by

$$q = f(k, l) = \alpha k + \beta l. \quad (9.36)$$

It is easy to show that this production function exhibits constant returns to scale: For any $t > 1$,

$$f(tk, tl) = \alpha tk + \beta tl = t(\alpha k + \beta l) = tf(k, l). \quad (9.37)$$

All isoquants for this production function are parallel straight lines with slope $-\beta/\alpha$. Such an isoquant map is pictured in Figure 9.4a. Because the *RTS* is constant along any straight-line isoquant, the denominator in the definition of σ (Equation 9.35) is equal to 0 and hence σ is infinite. Although this linear production function is a useful example, it is rarely encountered in practice because few production processes are characterized by such ease of substitution. Indeed, in this case, capital and labor can be thought of as perfect substitutes for each other. An industry characterized by such a production function could use *only* capital or *only* labor, depending on these inputs' prices. It is hard to envision such a production process: Every machine needs someone to press its buttons, and every laborer requires some capital equipment, however modest.

Case 2: Fixed proportions ($\sigma = 0$)

Production functions characterized by $\sigma = 0$ have L-shaped isoquants as depicted in Figure 9.4b. At the corner of an L-shaped isoquant, a negligible increase in k/l causes an infinite increase in *RTS* because the isoquant changes suddenly from horizontal to vertical there. Substituting 0 for the change in k/l in the numerator of the formula for σ in Equation 9.33 and infinity for the change in *RTS* in the denominator implies $\sigma = 0$. A firm would always operate at the corner of an isoquant. Operating anywhere else is inefficient because the same output could be produced with fewer inputs by moving along the isoquant toward the corner.

As drawn in Figure 9.4, the corners of the isoquants all lie along the same ray from the origin. This illustrates the important special case of a *fixed-proportions production function*. Because the firm always operates at the corner of some isoquant, and all isoquants line up along the same ray, it must be the case that the firm uses inputs in the fixed proportions given by the slope of this ray regardless of how much it produces.⁸ The inputs are perfect complements in that, starting from the fixed proportion, an increase in one input is useless unless the other is increased as well.

The mathematical form of the fixed-proportions production function is given by

$$q = \min(\alpha k, \beta l), \quad \alpha, \beta > 0, \quad (9.38)$$

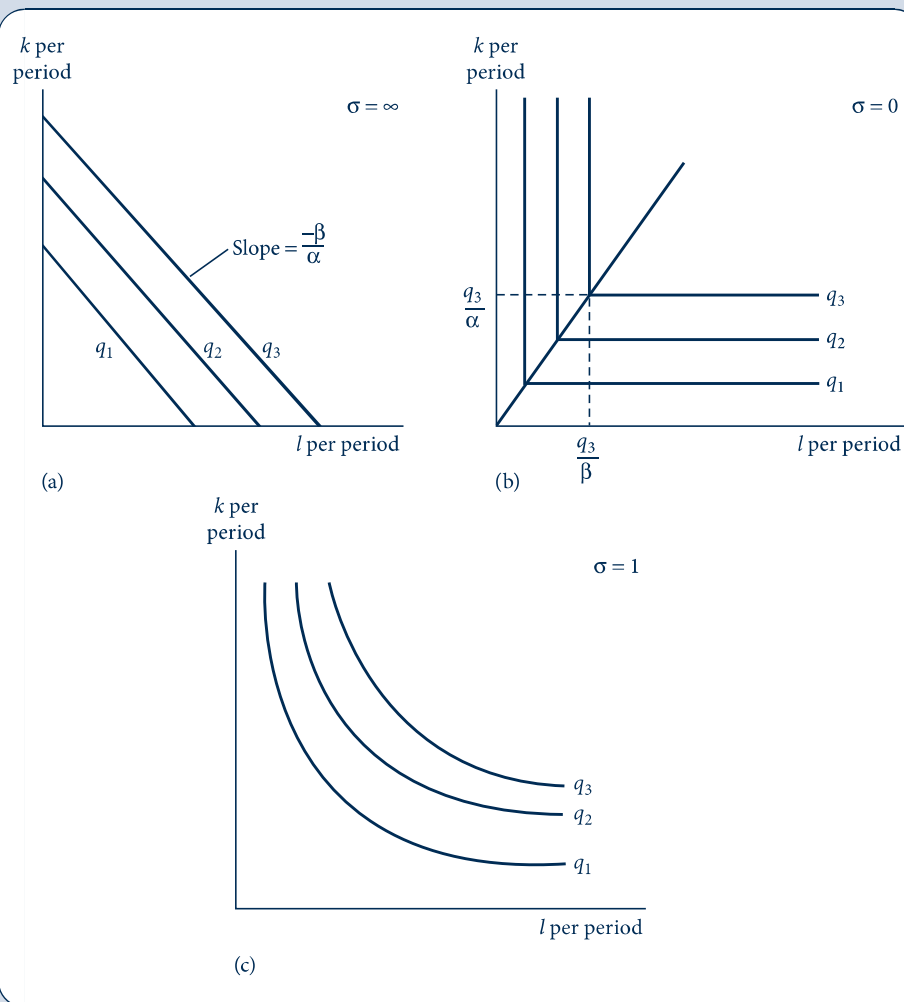
where the operator “min” means that q is given by the smaller of the two values in parentheses. For example, suppose that $\alpha k < \beta l$; then $q = \alpha k$, and we would say that capital is the binding constraint in this production process. The employment of more labor would

⁸Production functions with $\sigma = 0$ need not be fixed proportions. The other possibility is that the corners of the isoquants lie along a nonlinear curve from the origin rather than lining up along a ray.

FIGURE 9.4

Isoquant Maps for
Simple Production
Functions with Various
Values for σ

Three possible values for the elasticity of substitution are illustrated in these figures. In (a), capital and labor are perfect substitutes. In this case, the *RTS* will not change as the capital–labor ratio changes. In (b), the fixed–proportions case, no substitution is possible. The capital–labor ratio is fixed at β/α . A case of limited substitutability is illustrated in (c).



not increase output, and hence the marginal product of labor is zero; additional labor is superfluous in this case. Similarly, if $\alpha k > \beta l$, then labor is the binding constraint on output, and additional capital is superfluous. When $\alpha k = \beta l$, both inputs are fully utilized. When this happens, $k/l = \beta/\alpha$, and production takes place at a vertex on the isoquant map. If both inputs are costly, this is the only cost-minimizing place to operate. The locus of all such vertices is a straight line through the origin with a slope given by β/α .⁹

⁹With the form reflected by Equation 9.38, the fixed-proportions production function exhibits constant returns to scale because

$$f(tk, tl) = \min(\alpha tk, \beta tl) = t \cdot \min(\alpha k, \beta l) = tf(k, l)$$

for any $t > 1$. As before, increasing or decreasing returns can be easily incorporated into the functions by using a nonlinear transformation of this functional form—such as $[f(k, l)]^\gamma$, where γ may be greater than or less than 1.

The fixed-proportions production function has a wide range of applications. Many machines, for example, require a certain number of people to run them, but any excess labor is superfluous. Consider combining capital (a lawn mower) and labor to mow a lawn. It will always take one person to run the mower, and either input without the other is not able to produce any output at all. It may be that many machines are of this type and require a fixed complement of workers per machine.¹⁰

Case 3: Cobb–Douglas ($\sigma = 1$)

The production function for which $\sigma = 1$, called a *Cobb–Douglas production function*,¹¹ provides a middle ground between the two polar cases previously discussed. Isoquants for the Cobb–Douglas case have the “normal” convex shape and are shown in Figure 9.4c. The mathematical form of the Cobb–Douglas production function is given by

$$q = f(k, l) = Ak^\alpha l^\beta, \quad (9.39)$$

where A , α , and β are all positive constants.

The Cobb–Douglas function can exhibit any degree of returns to scale, depending on the values of α and β . Suppose all inputs were increased by a factor of t . Then

$$\begin{aligned} f(tk, tl) &= A(tk)^\alpha (tl)^\beta = At^{\alpha+\beta} k^\alpha l^\beta \\ &= t^{\alpha+\beta} f(k, l). \end{aligned} \quad (9.40)$$

Hence if $\alpha + \beta = 1$, the Cobb–Douglas function exhibits constant returns to scale because output also increases by a factor of t . If $\alpha + \beta > 1$, then the function exhibits increasing returns to scale, whereas $\alpha + \beta < 1$ corresponds to the decreasing returns-to-scale case. It is a simple matter to show that the elasticity of substitution is 1 for the Cobb–Douglas function.¹² This fact has led researchers to use the constant returns-to-scale version of the function for a general description of aggregate production relationships in many countries.

The Cobb–Douglas function has also proved to be useful in many applications because it is linear in logarithms:

$$\ln q = \ln A + \alpha \ln k + \beta \ln l. \quad (9.41)$$

The constant α is then the elasticity of output with respect to capital input, and β is the elasticity of output with respect to labor input.¹³ These constants can sometimes be

¹⁰The lawn mower example points up another possibility, however. Presumably there is some leeway in choosing what size of lawn mower to buy. Hence before the actual purchase, the capital–labor ratio in lawn mowing can be considered variable: Any device, from a pair of clippers to a gang mower, might be chosen. Once the mower is purchased, however, the capital–labor ratio becomes fixed.

¹¹Named after C. W. Cobb and P. H. Douglas. See P. H. Douglas, *The Theory of Wages* (New York: Macmillan Co., 1934), pp. 132–35.

¹²For the Cobb–Douglas function,

$$RTS = \frac{f_l}{f_k} = \frac{\beta Ak^\alpha l^{\beta-1}}{\alpha Ak^{\alpha-1} l^\beta} = \frac{\beta k}{\alpha l}$$

or

$$\ln RTS = \ln(\beta/\alpha) + \ln(k/l).$$

Hence

$$\sigma = \frac{\partial \ln k/l}{\partial \ln RTS} = 1.$$

¹³See Problem 9.5.

estimated from actual data, and such estimates may be used to measure returns to scale (by examining the sum $\alpha + \beta$) and for other purposes.

Case 4: CES production function

A functional form that incorporates all three previous cases and allows σ to take on other values as well is the constant elasticity of substitution (CES) production function first introduced by Arrow et al. in 1961.¹⁴ This function is given by

$$q = f(k, l) = [k^\rho + l^\rho]^{\gamma/\rho} \quad (9.42)$$

for $\rho \leq 1$, $\rho \neq 0$, and $\gamma > 0$. This function closely resembles the CES utility function discussed in Chapter 3, although now we have added the exponent γ/ρ to permit explicit introduction of returns-to-scale factors. For $\gamma > 1$ the function exhibits increasing returns to scale, whereas for $\gamma < 1$ it exhibits decreasing returns.

Direct application of the definition of σ to this function¹⁵ gives the important result that

$$\sigma = \frac{1}{1 - \rho}. \quad (9.43)$$

Hence the linear, fixed-proportions, and Cobb–Douglas cases correspond to $\rho = 1$, $\rho = -\infty$, and $\rho = -0$, respectively. Proof of this result for the fixed-proportions and Cobb–Douglas cases requires a limit argument.

Often the CES function is used with a distributional weight, α ($0 \leq \alpha \leq 1$), to indicate the relative significance of the inputs:

$$q = f(k, l) = [\alpha k^\rho + (1 - \alpha)l^\rho]^{\gamma/\rho}. \quad (9.44)$$

With constant returns to scale and $\rho = 0$, this function converges to the Cobb–Douglas form

$$q = f(k, l) = k^\alpha l^{1-\alpha}. \quad (9.45)$$

EXAMPLE 9.3 A Generalized Leontief Production Function

Suppose that the production function for a good is given by

$$q = f(k, l) = k + l + 2\sqrt{k \cdot l}. \quad (9.46)$$

¹⁴K. J. Arrow, H. B. Chenery, B. S. Minhas, and R. M. Solow, “Capital–Labor Substitution and Economic Efficiency,” *Review of Economics and Statistics* (August 1961): 225–50.

¹⁵For the CES function we have

$$RTS = \frac{f_l}{f_k} = \frac{(\gamma/\rho) \cdot q^{(\gamma-\rho)/\gamma} \cdot \rho l^{\rho-1}}{(\gamma/\rho) \cdot q^{(\gamma-\rho)/\gamma} \cdot \rho k^{\rho-1}} = \left(\frac{l}{k}\right)^{\rho-1} = \left(\frac{k}{l}\right)^{1-\rho}.$$

Applying the definition of the elasticity of substitution then yields

$$\sigma = \frac{\partial \ln(k/l)}{\partial \ln RTS} = \frac{1}{1 - \rho}.$$

Notice in this computation that the factor ρ cancels out of the marginal productivity functions, thereby ensuring that these marginal productivities are positive even when ρ is negative (as it is in many cases). This explains why ρ appears in two different places in the definition of the CES function.

This function is a special case of a class of functions named for the Russian-American economist Wassily Leontief.¹⁶ The function clearly exhibits constant returns to scale because

$$f(tk, tl) = tk + tl + 2t\sqrt{kl} = tf(k, l). \quad (9.47)$$

Marginal productivities for the Leontief function are

$$\begin{aligned} f_k &= 1 + (k/l)^{-0.5}, \\ f_l &= 1 + (k/l)^{0.5}. \end{aligned} \quad (9.48)$$

Hence marginal productivities are positive and diminishing. As would be expected (because this function exhibits constant returns to scale), the *RTS* here depends only on the ratio of the two inputs

$$RTS = \frac{f_l}{f_k} = \frac{1 + (k/l)^{0.5}}{1 + (k/l)^{-0.5}}. \quad (9.49)$$

This *RTS* diminishes as k/l falls, so the isoquants have the usual convex shape.

There are two ways you might calculate the elasticity of substitution for this production function. First, you might notice that in this special case the function can be factored as

$$q = k + l + 2\sqrt{kl} = (\sqrt{k} + \sqrt{l})^2 = (k^{0.5} + l^{0.5})^2, \quad (9.50)$$

which makes clear that this function has a CES form with $\rho = 0.5$ and $\gamma = 1$. Hence the elasticity of substitution here is $\sigma = 1/(1 - \rho) = 2$.

Of course, in most cases it is not possible to do such a simple factorization. A more exhaustive approach is to apply the definition of the elasticity of substitution given in footnote 6 of this chapter:

$$\begin{aligned} \sigma &= \frac{f_k f_l}{f \cdot f_{kl}} = \frac{[1 + (k/l)^{0.5}][1 + (k/l)^{-0.5}]}{q \cdot (0.5/\sqrt{kl})} \\ &= \frac{2 + (k/l)^{0.5} + (k/l)^{-0.5}}{1 + 0.5(k/l)^{0.5} + 0.5(k/l)^{-0.5}} = 2. \end{aligned} \quad (9.51)$$

Notice that in this calculation the input ratio (k/l) drops out, leaving a simple result. In other applications, one might doubt that such a fortuitous result would occur and hence doubt that the elasticity of substitution is constant along an isoquant (see Problem 9.7). But here the result that $\sigma = 2$ is intuitively reasonable because that value represents a compromise between the elasticity of substitution for this production function's linear part ($q = k + l$, $\sigma = \infty$) and its Cobb–Douglas part ($q = 2k^{0.5}l^{0.5}$, $\sigma = 1$).

QUERY: What can you learn about this production function by graphing the $q = 4$ isoquant? Why does this function generalize the fixed-proportions case?

TECHNICAL PROGRESS

Methods of production improve over time, and it is important to be able to capture these improvements with the production function concept. A simplified view of such progress is provided by Figure 9.5. Initially, isoquant q_0 records those combinations of capital and labor that can be used to produce an output level of q_0 . Following the development of superior production techniques, this isoquant shifts to q'_0 . Now the same level of output

¹⁶Leontief was a pioneer in the development of input-output analysis. In input-output analysis, production is assumed to take place with a fixed-proportions technology. The Leontief production function generalizes the fixed-proportions case. For more details see the discussion of Leontief production functions in the Extensions to this chapter.